

THE KCP LIMITED

CEMENT UNIT-II, RAMAKRISHNAPURAM

OUR AIM : ZERO HARM



PRESENTATION ON “EMERGENCY PREPAREDNESS”



Why Emergency Preparedness

- To take prompt corrective actions to save lives and sufferings/injuries.
- To protect plant machines, equipment & buildings etc.
- To protect loss of materials.
- To Protect and minimize property losses.
- To resume normal situation or operation within minimum possible time.



NEED OF EMERGENCY PLAN

- AFTER BHOPAL GAS DISASTER IN 1984, THE FACTORIES ACT 1948 WAS AMENDED IN 1987 AND NEW PROVISIONS REGARDING SAFETY AND EMERGENCY MANAGEMENT WERE ADDED. (On 3 December 1984, over 500,000 people in the vicinity of the [Union Carbide India Limited](#) pesticide plant in [Bhopal](#), [Madhya Pradesh](#), [India](#) were exposed to the highly [toxic](#) gas [methyl isocyanate](#), in what is considered the [world's worst industrial disaster](#).)
- NOW IT IS A MANDATORY TO AN ORGANISATION TO PREPARE EMERGENCY PREPAREDNESS PLAN HANDLING HAZARDOUS CHEMICALS AS PER MSIHC RULES 1989.
- UNDER THE PROVISION OF SECTION 41B (4) OF FACTORIES ACT 1987, EVERY OCCUPIER SHALL HAVE VALID EMERGENCY PLAN FOR HIS FACTORY.



LG Polymers

The term "AP DISASTER GAS PLANT" refers to the [Visakhapatnam gas leak](#) that occurred on May 7, 2020, at the LG Polymers India plant in Andhra Pradesh, India. The incident involved a large-scale release of [styrene gas](#), a toxic chemical, from a storage tank while the plant was being re-opened after a coronavirus-related lockdown. The leak resulted in multiple deaths, widespread injuries, and significant environmental impact on the surrounding communities.

- **Date:** May 7, 2020.
- **Location:** RR Venkatapuram village, near Visakhapatnam, Andhra Pradesh, India.
- **Company:** LG Polymers India Pvt Ltd.
- **Chemical:** Styrene gas.
- **Cause:** Leakage from a storage tank as the plant was being re-started.
- **Impact:**
 - **Fatalities:** 12 people died.
 - **Injuries:** Hundreds of people were hospitalized and affected by the toxic gas.
 - **Environment:** A vapor cloud spread over a radius of approximately three kilometers, impacting nearby villages and the environment.

- **Lessons Learned:**

- The incident highlighted the need for improved industrial safety practices, stricter enforcement of safety regulations, and lessons from the Bhopal gas tragedy to prevent future industrial disasters.



7 killed, 20 injured in reactor blast at pharma company in Andhra Pradesh's Anakapalli

The death toll is likely to go up, as bodies of several workers were feared trapped under the rubble of the slab of the first floor of the four-storey building; as many as 381 workers have been working in two shifts in the company and the accident occurred during shift change.

At least 17 workers were killed and 20 others sustained burns in a major fire that broke out reportedly after a reactor blast in Escientia Advanced Sciences Private Ltd. in the Special Economic Zone at Atchutapuram in Anakapalli of Andhra Pradesh on Wednesday. The company manufactures intermediate chemicals and pharmaceutical ingredients.



SCOPE

- **EMERGENCY PLAN COMES IN TO EFFECT AT THE DECLARATION OF EMERGENCY AND EXTENDS UP TO TERMINATION OF THE EMERGENCY.**
- **IT IS APPLICABLE TO ALL EMPLOYEES AND PERSONS AVAILABLE IN THE PLANT/FACILITY.**



Emergencies

- **ON SITE EMERGENCY**

Controlling Authority is Unit Head

- **OFF SITE EMERGENCY**

Controlling Authority is District Collector.



Objectives of the Emergency Plan

- This plan is a standing instructions for handling 'Emergency Situations.
- The plan envisages pooling of resources to mitigate emergency.
- To safe guard human beings, machines, equipment and property.
- To safeguard other persons not affected by emergency so far.
- Identify/rescue the casualties and arrange for their treatment.
- To define channels of communications with key personnel.



Contents of Emergency Plan

The emergency plan for industrial facilities should include:

- Brief Process description
- Various hazards available
- Classification of emergencies
- Emergency scenario envisaged
- Agencies required for Emergency handling
- Declaration and notification of emergency
- Emergency Control centre and assembly areas
- Communication facilities available
- First-aid centre and hospital facilities



Cont.,

- Responsibilities of key personnel/Alternate persons with contact numbers.
- Monitoring of hazardous chemicals.
- Fire fighting and rescue arrangements.
- Assessment of conditions subsequent to release of toxic chemicals, rescue.
- Termination of emergency and notification of the same.
- Maintenance of emergency equipment.
- Surveillance and record keeping.
- Mutual aid scheme members and communication system among members.
- Periodicity of conducting drills/exercises.



ACCIDENT SCENARIOS

- Massive leakage of toxic chemicals from process piping / equipment.
- Catastrophic failure of vessel, piping of chemical.
- Major fire
- Leakage of a road tanker containing hazardous chemicals likely to affect our site.
- Severe lightning / storm condition / flooding of plant area due to excessive rain.
- Severe earth quake and / or any explosion near the plant
- Air craft or missile impact on / near the Plant.



Declaration of emergency

- Commence :- Short intermittent signals 5 seconds 'ON' & 5 Seconds 'OFF' for a total period of 3 minutes.
- Termination :- A continuous signal (All clear) for 3 minutes.
- Emergency announcement

Emergency announcement on PA System shall be made in English, & Telugu language.



RESOURCES

- Communication systems (including Public Address System)
- Emergency Siren
- Transport for evacuation of plant personnel
- Assembly area
- First Aid facility including ambulance at site
- Fire Fighting and rescue arrangements
- Security arrangements
- Breathing air sets and facilities for bottling of breathing air.
- Laboratory facilities
- Fire Alarm System –Heat & Smoke Detectors
- Fire Pump House



Types of Emergencies

Potential Incidents leading to environmental damage, serious injuries or ill health to personnel identified in our organization are as follows:-



Types of Emergencies

S.No	Type of Emergencies	Possible Occurrence of Emergencies	Applicability to KCPL		Expertise of KCPL to handle/ Manage/Mitigate
			Yes	No	
01	Electrical Short Circuit Incident/Electrical Shock	Possible	Yes		Very Good
02	Chemical Fire Incident	Remote	Yes		Satisfactory
03	Machine – Operation-Fatal/ Major Injury to Personnel	Possible	Yes		Excellent
04	Fire due to diesel in Storage Area	Possible	Yes		Very Good
05	Asphyxiation	Remote	Yes		Excellent
06	High Level of Dust	Possible	Yes		Good
07	Natural disasters(Earth Quake/Floods/earth cave-in)	Possible	Yes	No	Poor
08	Bad weather(cyclones/heavy rains/wind)	Possible	Yes		Satisfactory
09	Loss/disturbance of utility supply(e.g. loss of electric power/ short circuit /earth fault on incoming transmission line /Sudden failure of electrical equipment)	Remote	Yes		Good



Types of Emergencies

10	Pandemics/epidemics/outbreaks of communicable disease,	Possible	Yes		Poor
11	Civil disturbance	Possible		No	Poor
12	Terrorism ,Bomb Threat & Sabotage	Remote		No	Poor
13	Workplace violence,	Possible	Yes		Good
14	Failure of critical equipment,	Possible	Yes		Very Good
15	Traffic accidents.	Possible	Yes		Very Good
16	Abnormal conditions such as Start-up/shut down/construction/demolition activities.	Possible	Yes		Very Good
17	Failure of Plant Automation System	Possible	Yes		Very Good
18	Structural Collapse	Remote	Yes		Good
19	Fire from Waste oil storage	Possible	Yes		Very Good
20	Toppling of tankers/trucks within plant premises	Possible	Yes		Good
21	Spillage/ leakage of Waste oil	Occasional	Yes		Good
22	Oil spills in the storm water drains	Occasional	Yes		Very Good



Types of Emergencies

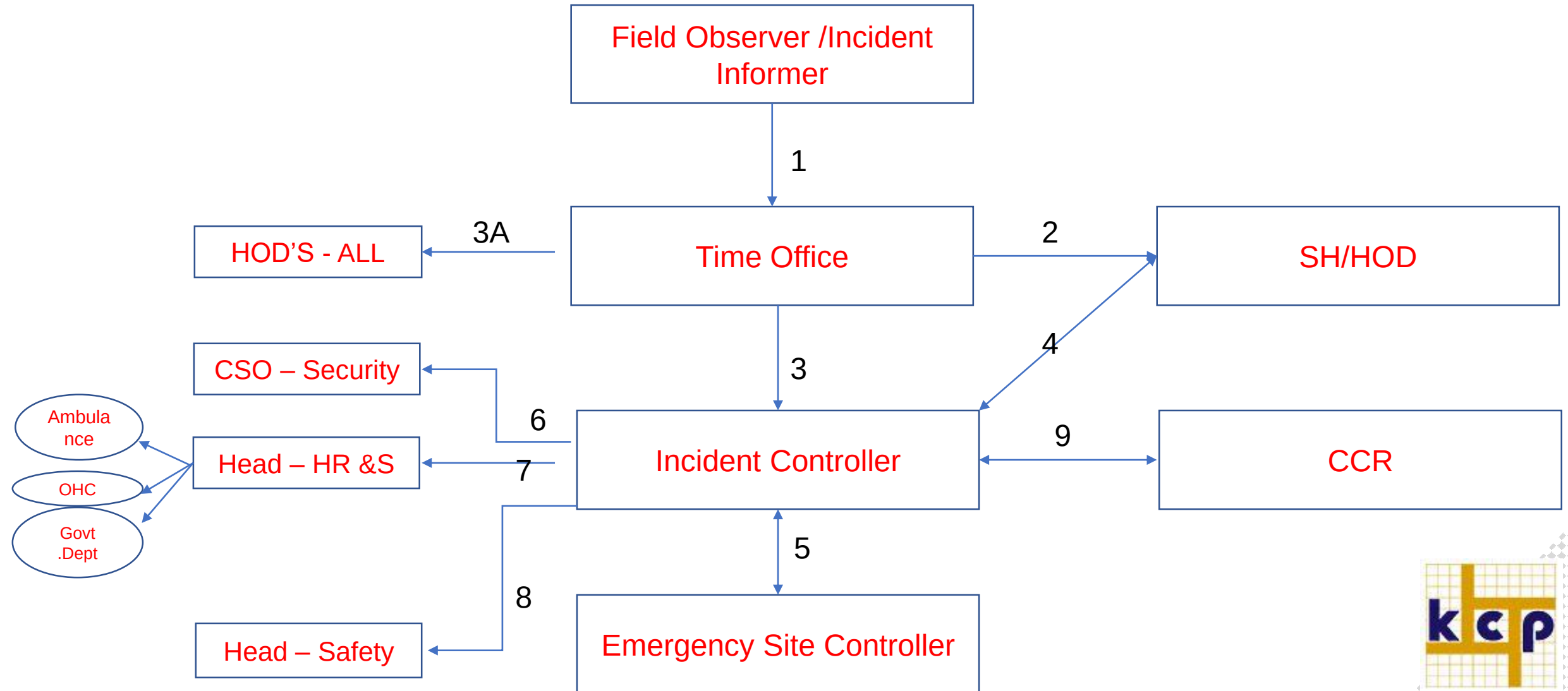
23	Explosion of Pressure Vessel	Possible	Yes		Very Good
24	Fire from the Raw Coal and Fine Coal	Possible	Yes		Very Good
25	Fire from the Bag House and ESP	Possible	Yes		Excellent
26	Explosion in the Bag House	Possible	Yes		Excellent
27	Fire &/or Explosive from Explosive Storage	Possible	Yes		Excellent
28	Food / Water poisoning	Possible	Yes		Very Good
29	Animal Bite(Snake /Bee /Dog/Monkey/Insect)	Possible	Yes		Very Good
30	Cyclone Jam	Possible	Yes		Very Good
31	Hot Material (Clinker & Ground Lime Stone) Spillage	Possible	Yes		Very Good
32	Fire from the Conveyor Belts	Possible	Yes		Very Good
33	Silo Collapse and Fire dust spread every where	Possible	Yes		Very Good



Emergency Communication Flow chart



EMERGENCY COMMUNICATION –FLOW DIAGRAM



EMERGENCY SIREN CODE

అత్యవసర సైరన్ సూచిక



**INTERRUPTED WAILING SOUND
OF 45 SECONDS DURATION
WITH A GAP OF 15 SECONDS**

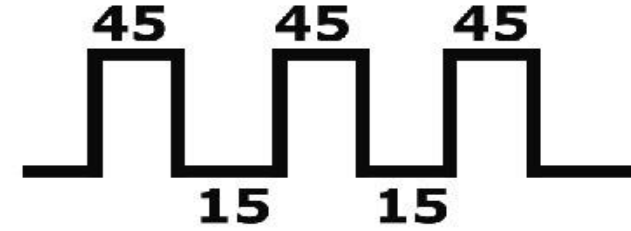
15 సెకండ్ల అంతరముతో కూడిన

45 సెకండ్ల చీర్లు ధ్వని

**3 MINUTES CONTINUOUS BLAST
WILL DENOTE ALL CLEAR &
RESTORATION OF NORMALLY**

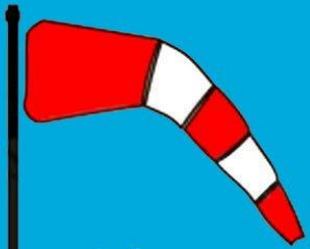
**3 నిమిషాల నిరంతర ధ్వని శబ్దంతో
సామాన్య స్థితి ఏర్పడుటను సూచిస్తుంది**

LONG & SHORT BLAST

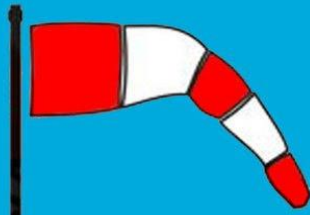


ALL CLEAR SIGNAL

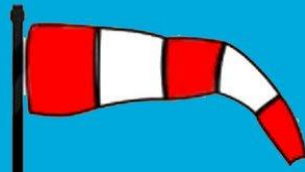




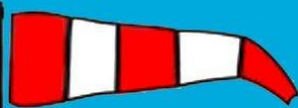
3 knots
5.5 km/h
3.5 mph



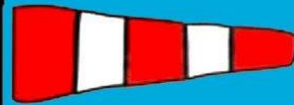
6 knots
11 km/h
7 mph



9 knots
16 km/h
10 mph



12 knots
22 km/h
14 mph



15+ knots
28+ km/h
17+ mph

HOW TO READ A WIND CONE?



aviationrenewables

A **windsock** (also known as **wind cone** or **wind sleeve**) is a [conical](#) textile tube that resembles a giant [sock](#). It can be used as a basic indicator of [wind speed](#) and [direction](#),

Locations:

1. Time Office
2. CPP



Details of Emergency Assembly Points & Locations



TIME OFFICE

CCR

IN FRONT PUMP HOUSE

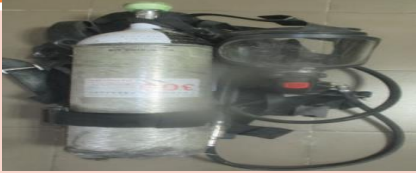
CPP MAINGATE



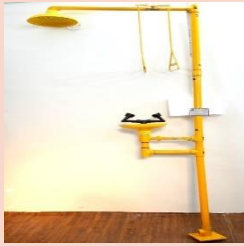
EMERGENCY MITIGATING EQUIPMENTS

S.NO	EQUIPMENTS	PHOTO
1	FIRE TENDER	
2	FIRE EXTINGUISHERS	
3	NO. OF FIRE HYDRANT POINTS	
4	WATER MIST	
5	SMOKE DETECTORS	

EMERGENCY MITIGATING EQUIPMENTS

S.NO	EQUIPMENTS	PHOTO
6	SELF CONTAINED BREATHING APPARATUS	
7	EMERGENCY ASSEMBLY POINTS	
8	EMERGENCY SIREN	
9	AMBULANCE	
10	FIRST AID BOXES	

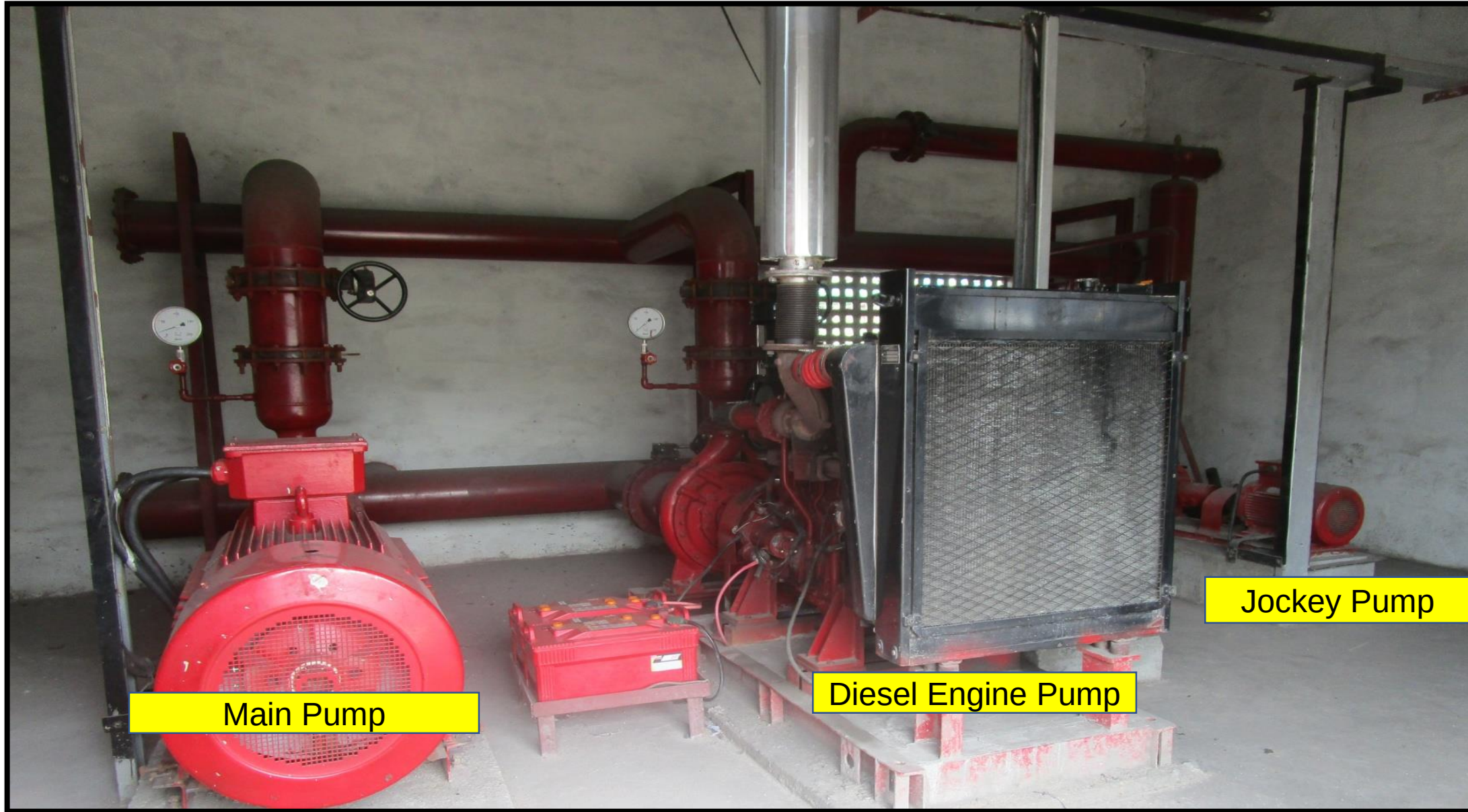
EMERGENCY MITIGATING EQUIPMENTS

S.N O	EQUIPMENTS		PHOTO
11	EYE WASH & BODY WASH FOUNTAIN		
12	SURVEILLANCE CAMERAS		
13	FIRE HYDRANT SYSTEM		

FIRE HYDRANT LINE



FIRE HYDRANT SYSTEM



Main Pump

Diesel Engine Pump

Jockey Pump

Fire Hydrant System Pumps Details

PUMP DETAILS				
Sr.No	Pump Type	Capacity	Auto Start	Auto Stop/ Manual Stop
01	Jockey Pump	10.8 m3/hr	7kg/cm2	8kg/cm2
02	Elect.Main Pump	273 m3/hr	5kg/cm2	Manual Stop
03	Di esel Pump	273 m3/hr	4kg/cm2	Maannual Stop

Water Tank Capacity : 2025 Cu.Mtr

Diesel Tank Capacity : 200 Ltrs.

ACTION PROCEDURE DURING EMERGENCY

- The Site Controller is the Authority for declaring the Emergency. The field operator or any other person noticing the emergency shall inform to the Time Office first.
- The Time office employee will inform respective SIC/ HOD, All HOD, Security, First Aid Team and Rescue team through Phone.
- Then the security inform to the Emergency controller immediately. The SIC/ HOD will rush to the spot, assess the situation and inform to Emergency Controller, then the Emergency Controller after discussion with Chief Emergency controller will take a decision to declare emergency.
- The declaration of major emergency puts many agencies on action & regular system may be disturbed which may be very costly at times or the emergencies may be serious. Hence the Site Controller will only declare the emergency after assessing the situation through Emergency Controller and Safety coordinator.



Emergency Activities

- Emergency Situation to be communicated to Time office
- CCR operator will raise the Emergency alarm with concurrence from concerned authority (Emergency controller)
- On hearing the siren all employees except CCR and other control room operators shall stop their work, keep the things safe and rush to Safe Assembly point.
- Security personnel will help in evacuating the people from plant area to Assembly point.
- The Emergency controller has come to Emergency control centre from there he will give instruction to all and bring the situation under his control.
- Security / HR shall reach the Assembly point and start head count.
- After head count, report to be communicated to Emergency controller.
- Emergency controller will instruct the task force to search the missing or affected person.
- Then Ambulance will be sent to search place and bring the affected victims to OHC for treatment.
- Security Officer and Safety officer will give direction to fire fighting team to extinguish the fire.
- Once the fire is put off, and area is clear, the fire officer will inform the status to Emergency controller.
- The Emergency controller will ensure the same with Chief Emergency controller and give direction to CCR personal for all clear siren.
- The Environment Officer will analyse the impact on environment
- Once the employees hear all clear siren, they should go to their work area.
- Post meeting will be called by Safety dept. head and review for the improvements



EMERGENCY CONTROL CENTRE

The Emergency Control Centre is located at the CCR Ground floor . It is the place from which the operations to handle the emergency are directed and coordinated. It will be attended by the Chief Emergency Controller, Emergency Controller and other key personnel.

The Centre is equipped to receive and transmit direction and information from and to the Emergency Controller and plant areas as well as outside. It is sited in an area of minimum risk and close to the main road for easy access.

In addition to the means of communication, the Centre is equipped with relevant data and equipment which will assist those manning the Centre to be conversant with that developing situation and enable to plan accordingly.

The Emergency Control Centre is equipped with the following:

- 1)Telephone (Intercom and P&T).
- 2)Telephone Directory (Internal & External)
- 3)Plot Plan of the Factory with sufficient details clearly marked.
- 4) Note pads, pens, pencils, rubber and stationery to record all messages received and sent by what so ever means.
- 5) Copies of the On Site Emergency Plan
- 6)Emergency Safety Cupboard containing essential PPE.



EMERGENCY CONTROLLER

- Assess the scale of emergency and decide if a major emergency exists or is likely. On his decision he will activate the Onsite Emergency Plan
- On Consultation with Site controller ask CCR operator to blow siren & declare Emergency.
- Direct the shutting down and evacuation of plant and areas likely to be affected by the emergency.
- Ensure that the outside emergency services, including the mutual aid have been called in.
- Ensure that the key personnel have been called in.
- Direct all the operations within the affected area with the following priorities:
 - a) Secure the safety of personnel.
 - b) Minimise damage to plant, property and environment.
 - c) Minimise loss of material.
- Direct rescue and Fire fighting operations until the arrival of outside fire brigade.
- Search for casualties (if any). Evacuate non essential workers to the assembly point.
- Give advice and information as required to the heads of fire brigade and other emergency services.
- Brief the Site Main Controller and keep informed of developments.
- Preserve evidences that will be necessary for subsequent inquiry into the cause of the emergency and concluding preventive measures.



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Security Coordinator:

On receipt of the information immediately rush to the scene and he will

- Assume full responsibility in organizing Fire fighting and security measures to mitigate the damage to life and property.
- Provide latest information to the Emergency Controller.
- Depute one Sr. Security for assisting the Emergency Controller.
- Co-ordinate with the external fire service as and when they arrive and direct them as required.
- Close all the gates immediately and restrict entry & exit to and from the plant.
- All emergency vehicles, like, ambulance, fire service vehicles, additional support vehicles, etc. shall only be allowed to enter the plant or go out of the plant area.
- No person, unless those authorized by the Incident Controller will be permitted to enter or leave the plant premises.
- The security officer will be responsible for controlling the movement of personnel.
- Coordinate with the police personnel if called for controlling the situation.
- Ensure that all the roads shall be kept clear for emergency and supporting vehicles' movement.
- Control Traffic inside factory & arrange alternate transport if need arises.
- Ensure that material movement activity also is restricted till termination of emergency.
- Start Head count immediately and verify the data of the present strength inside the plant area and inform the details to Incident Controller. The Incident Controller and the Shift In-charge (SIC) shall ensure that all personnel are rescued and head count tallied.
- The details of injured personnel being sent out and other personnel coming in shall be maintained at security posts and the same shall be accounted during the head count.



HR Coordinator

The Head HR & Personnel will act as Liaison officer on hearing the Fire Siren/Alarm, he will rush to the control center act as per the instruction of the Emergency Controller. His responsibilities are:

- Inform State Govt. Officials as per List.
- Essential officials of KCPL to be kept informed.
- Handle Police, Press and other enquiries
- Ensure the causality receive adequate attention
- Control Traffic inside factory arrange alternate transport if need arises.
- If emergency prolonged arrange for refreshment/catering



MEDICAL COORDINATOR

On hearing the emergency Siren/Alarm or Phone message, the Medical Officer will send First Aid team to report to the incident controller.

- Ambulance will be made available at the control centre.
- If the Ambulance leaves the control centre it should be informed to Emergency Controller.
- If the Ambulance leaves with patient the name and other particulars, Hospital, Destination etc. should be informed to the Emergency Controller.
- The Medical Attendant informs the Chief at the Hospital about the causality reaching Hospital for treatment.



REVIEW AND UPDATE OF EMERGENCY PLAN

The Emergency Plan will be reviewed and updated periodically at least once in two year. If any changes in the layout, machinery, plant or process should be updated in the plan.

The Plan is made Known to all employees including contract workers by imparting specific training. The adequacy of the plan and the training is assessed by conducting mock drills.



Thank You

